

Introduction

Project: Smart Library Request Workflow in ServiceNow

The Smart Library Request Workflow project aims to digitize and automate the end-to-end process of library resource requests using the ServiceNow platform. Traditionally, library management involves manual tracking of book requests, approvals, inventory checks, and returns — a process that is time-consuming and error-prone. This project leverages ServiceNow's workflow automation, case management, and catalog capabilities to build a smart, self-service portal where users can request books or resources, track approval status in real-time, and receive automated notifications at every stage. By integrating backend configurations, data migration from legacy systems, and robust security testing, the solution ensures a seamless, scalable, and secure library management experience for both administrators and end-users. The project is structured across six key epics — from initial requirement gathering to final deployment and conclusion — ensuring a systematic and well-documented delivery lifecycle.

Problem Statement

- In many academic institutions, managing a library system manually creates challenges such as lack of real-time updates on book availability, inefficient approval processes, and limited visibility into borrowing patterns. Students often face issues in knowing which books are available, while librarians struggle with tracking requests, approving or rejecting them, and maintaining accurate records of issued or returned books
- To address these challenges, a role-based Library Management System needs to be implemented on the ServiceNow platform. This system should allow students to request books, while librarians manage the book inventory and approve borrow requests. Automated workflows, role-based access... [text cuts off]

What are challenges faced without Smart Library Request Workflow in ServiceNow

- Manual book request and approval process.
- Time-consuming library operations.
- Difficulty in tracking book availability.
- Delays in request approval and book issuance.
- Paper-based record management leading to errors.
- No real-time status tracking for students.
- Lack of automated email notifications.

Objective

- The objective of this project is to implement a role-based Library Management System in ServiceNow that simplifies the process of borrowing and managing books. The system allows students to submit borrow requests and track their status, while librarians manage the book catalog and approve or reject requests. Through automated workflows, book availability is updated in real time, role-based access ensures proper security, and reporting features like “Most Borrowed Books” provide insights for better resource management.

Requirement Analysis

Solutions Requirement

TEAM ID	SWTID-2026-6704
PROJECTNAME	Smart Library Request Workflow in Servicenow

Functional Requirements:

Following are the functional requirements

FR NO.	Functional Requirements	Sub requirement
FR-1	Requirement Analysis & Planning	Business Objectiv, Functional Scope, Stakeholder Mapping Execution Roadmap

FR-2	Backend Development & Configurations	Creation of roles Create,Tables,Create Tables Add Fields to Book table Add Fields to Borrow Request table,Configure Reference Qualifier (PreventRequesting Already-Issued Books)
FR-5	Deployment, Documentation & Final Presentation	TroubleshootingAdherence to Timelines,Innovation Document Technical Blueprint .Document Setup Manual,Project Demo Video Planning, Visual Demonstration Explanation Clarity.

Non-Functional Requirements

Following are the non-functionalRequirement

FN NO	Non-functional Requirement	Description
NFR-1	Usability	The Smart Library Request Workflow provides a user-friendly and intuitive interface that allowsstudents, staff, and librarians to easily request, issue, return, and track library resources with minimaltraining or technical knowledge.
NFR-2	Security	The Smart Library Request Workflow ensures data protection through user authentication, role-based access controls, and secure encryption to prevent unauthorized access and safeguard library records.
NFR-3	Reliability	The Smart Library Request Workflow consistently performs its functions without failure, ensuring accurate book tracking,

NRF-4	Performance	timely status updates, and dependable system availability for all users.
NRF-5	Available	The Smart Library Request Workflow is accessible at all times, ensuring users can request and track library resources anytime without downtime or service interruptions.
NRF-6	Scalability	The Smart Library Request Workflow can efficiently handle growing numbers of users, books, and requests, making it adaptable to the expanding needs of any institution or library branch.

Data Flow Diagram

The proposed system begins when a user submits a book or resource request, which flows into the ServiceNow system where it is stored and routed for processing. From there, the request moves to the librarian approval stage, where the librarian or administrator reviews the request and either approves or rejects it based on organizational policy. Once approved, the system performs an availability check, validating in real time whether the requested resource is currently in the library's inventory.

User

Student or faculty member who raises book/resource requests.

Logs into the ServiceNow Service Portal.

Searches the library catalog for available books/resources.

Submits a request through the self-service catalog item.

Checks real-time availability before submitting.

Tracks request status (pending, approved, rejected, fulfilled).

Smart Library Request Workflow



Architecture of service now

- Requesters (students/faculty) log in via ServiceNow Service Portal.
- Requests raised through Service Catalog (cat_item, sc_request, sc_req_item).
- Custom table maintains book/resource details and real-time availability.
- Roles defined for Requester, Librarian, and Admin.
- ACLs restrict data visibility based on role.
- Flow Designer automates approval routing and task assignment.
- Librarian/Admin reviews and approves/rejects requests.

Project Design

Proposed Solution: Smart Library Request Workflow (ServiceNow)

SNO	Parameter	Description
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1	Problem Statement (problem to be solved)	Educational institutions and organizations often struggle to manage library operations efficiently. Book requests, issuing, renewals, and returns are still handled manually or through disconnected legacy systems, leading to delayed approvals, misplaced records, lack of real-time visibility into book availability, and difficulty tracking overdue items. This results in poor user experience for students and staff, and increased administrative burden for library staff
2	Idea / Solution Description	The proposed solution is a Smart Library Request Workflow built on the ServiceNow platform — a centralized, automated system that digitizes the entire library request lifecycle. Users can raise book requests through a self-service portal or Virtual Agent chat, requests are automatically routed for approval using Flow Designer, and the system tracks issuing, due dates, renewals, and returns in real time, with dashboards giving library staff complete operational visibility.
3	Novelty / Uniqueness	Unlike traditional library management tools, this solution combines conversational AI (Virtual Agent) for hands-free requests, no-code automated approval workflows via Flow Designer, proactive notification alerts for due dates and overdue books, and live analytics dashboards — delivering an intelligent, self-service experience that eliminates manual registers and reduces staff intervention.

4	Social Impact / Customer Satisfaction	By automating requests and approvals, the workflow significantly reduces waiting time for students and staff, increases transparency around book availability, minimizes the risk of lost or misplaced records, and encourages accountability through timely reminders — fostering a more responsible and efficient library culture across the institution.
5	Business Model (Revenue Model)	The solution follows a Software-as-a-Service (SaaS) subscription model, license to educational institutions, universities, and corporate libraries on a per-user or per institution basis. Tiered pricing plans can be offered — a Basic tier covering core request and tracking features, and a Premium tier adding advanced analytics, custom reporting, and integration support.
6	Scalability of the Solution	Built on ServiceNow's enterprise-grade cloud platform, the solution is inherently scalable — capable of growing from a single library branch to a multi-campus or multi-organization deployment. It can seamlessly handle increasing numbers of users, expanding book catalogs and higher request volumes while maintaining consistent performance and reliability.

Phase 1: Requirement Analysis & Planning

- The system requires two roles: student and librarian. Students can request and track books, while librarians manage the catalog and approve requests. Two tables are needed: *Book* (details, status) and *Borrow Request* (transactions). Workflows will automate approvals and status updates, ACLs will control access, and reports like “Most Borrowed Books” will give insights. Planning covers defining tables, roles, workflows, and test cases to ensure smooth implementation.

Business Objectives

The main business objective of this project is to create a digital library system in ServiceNow that reduces manual effort, improves accuracy in book tracking, and enhances user experience for both students and librarians. By automating borrow requests, approvals, and book status updates, the system ensures efficient operations, transparency, and faster access to resources. Additionally, role-based controls and reporting features support better decision-making and effective management of library resources.

Functional scope:

The Library Management System on ServiceNow will provide the following key functions.

Book Management – Librarians can add, update, and maintain book details with status tracking (Available, Issued, Lost, Returned).

Borrow Requests – Students can create and track requests; librarians can approve, reject, or close them.

Workflow Automation – Approval of requests will automatically update the book status.

Role-Based Access – Students have limited access to requests, while librarians manage the entire system.

Reporting – Generate insights such as “Most Borrowed Books” for library usage analysis.

Stakeholder Mapping

Students – Request and track books; expect easy access and quick approvals.

Librarians – Manage catalog, approve/reject requests, generate reports; need efficiency and accuracy.

System Administrators – Configure tables, workflows, and roles; ensure secure and stable systems.

Management/Faculty – Use reports for planning and resource allocation.

IT Support Team – Provide technical support, upgrades, and system reliability.

Execution Roadmap

Role Setup – Create student and librarian roles. **Table Design** – Build Book and Borrow Request tables with required fields.

Form Configuration – Add fields, set default values, and establish related lists.

Workflow Automation – Configure Flow Designer for approval and book status updates.

Access Control – Define ACLs for role-based permissions.

UI Enhancements – Apply UI Policies for form restrictions and usability.

Reporting – Create dashboards/reports like Most Borrowed Books.

Testing & Validation – Use sample data, impersonation, and debug tools to verify end-to-end flow.

Milestone 1: Create Roles

- Define roles: student and librarian.
- Ensure role separation for access control (students = request, librarians = manage).

Milestone 2: Create Tables Create a Book table (book).

- Create a Book table (u_book).
- Create a Borrow Request table (u_borrow_request).
- Link Borrow Request → Book.

Milestone 3: Add Fields to Book

- Add fields: Title, Author, ISBN, Status.
- Configure Status choices: Available (default), Issued, Lost.

Milestone 4: Add Fields to Borrow Request

- **Add fields:** Requested By, Book (Reference), Request Date, Status.
- Configure Status choices: Requested (default), Approved, Rejected, Returned.

Milestone 5 : Show Borrow Requests on Book (Related List)

- Add Borrow Request related list on Book form.
- Verify relationship: Book → Borrow Request.

Milestone 6: Prevent Requesting Already-Issued Books (Reference Qualifier)

- Configure Reference Qualifier on Borrow Request → Book field.

- Condition: Show only books where Status = Available.

Milestone 7: Flow Designer – Approve → Issue Book

Create Flow: Trigger when Borrow Request → Status changes to Approved.

- Action: Update Book → Status = Issued.
- Optional: Send email notification to requester.

Milestone 8: UI Policy

- Apply UI Policy: When Borrow Request Status = Approved → lock fields (Book, Requested By, etc.).
- Improves form usability and prevents accidental edits.
- Milestone 9: Access Control (ACLs)
- Define ACLs for Book table:
- Librarians = Create/Write/Delete.
- Students & Librarians = Read.
- Define ACLs for Borrow Request table:
- Students = Create/Read (own records).
- Librarians = Approve/Write/Delete.

Milestone 10: Report – “Most Borrowed Books”

- Build Report from Borrow Request table.
- Group by Book, Count requests.
- Display the top 5 most borrowed books as a chart or list.

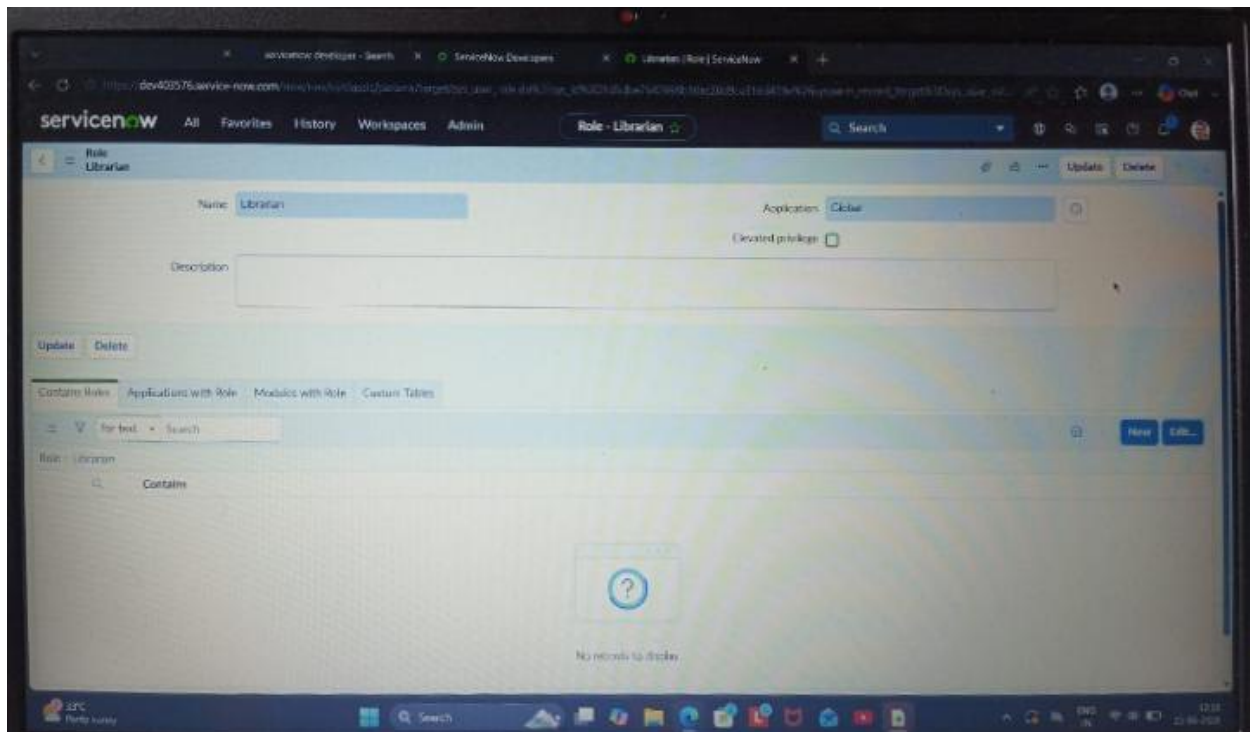
Phase Backend Development & Configuration

Data Architecture

Creations of Roles

1. Open service now
2. Click on All → search for Roles.
3. Select User Administration → Roles.
4. Click on New.

5.

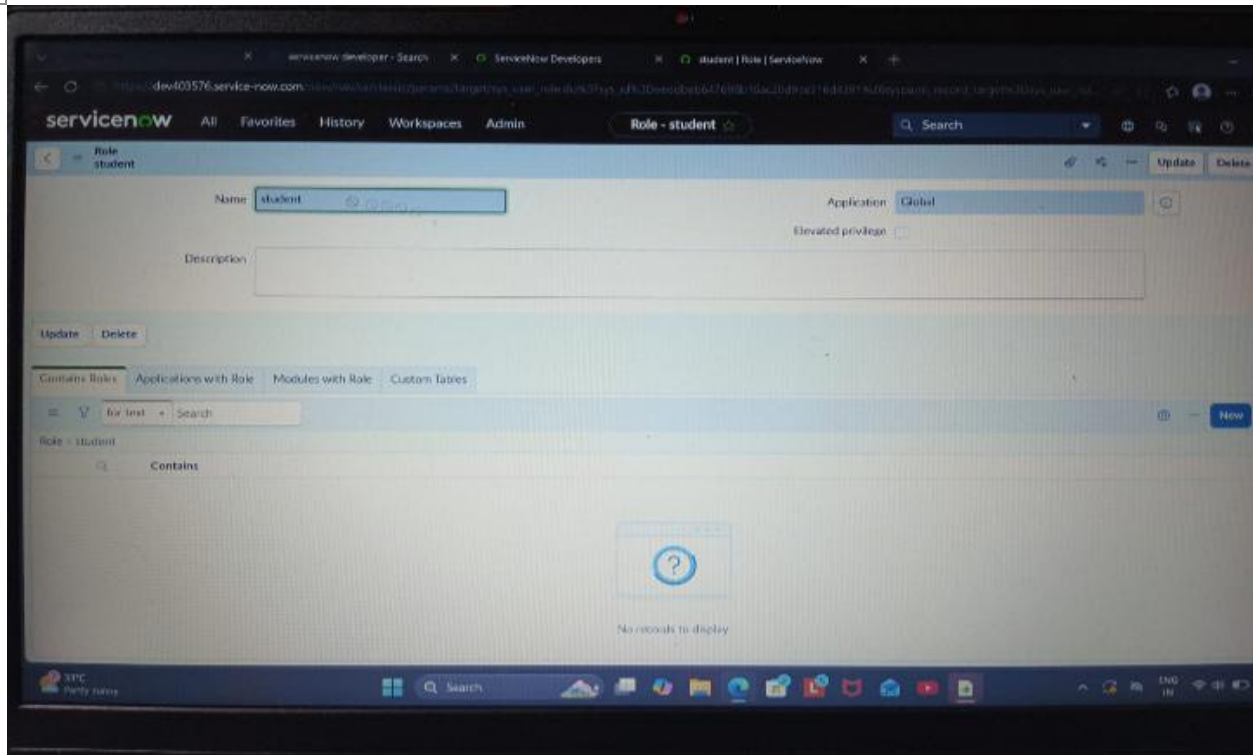


Enter RoleName as librarian → click Submit.

6. Again, click on New.

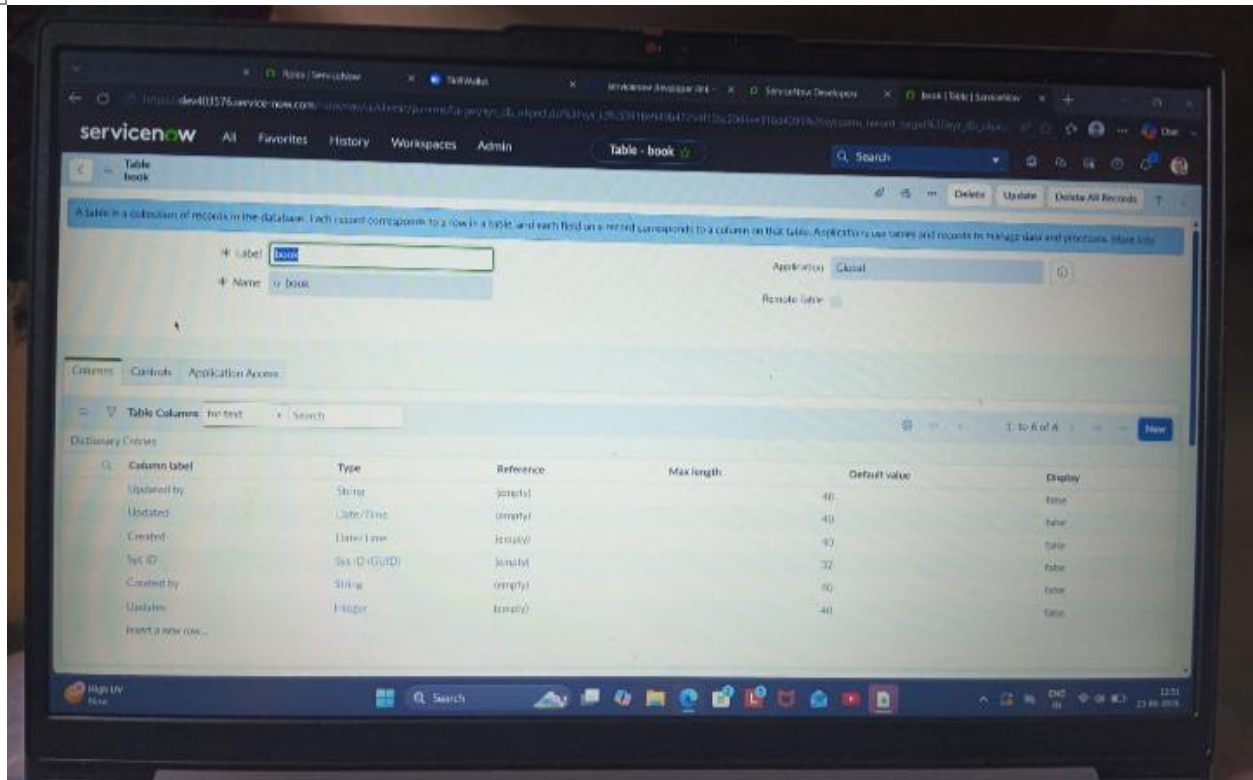
7. Enter RoleName as student → click Submit

8. Verify that both roles (librarian and student) are listed in the Roles module.



Create table

1. Open ServiceNow.
2. Click on All → search for Tables.
3. Select System Definition → Tables.
4. Click on New.
5. Enter Label as Book → click Submit (this creates the tablebook).



6. Again, click on New.

7. Enter Label as Borrow Request → click Submit (this creates `tblu_borrow_request`).

8. Verify that both tables (Book and Borrow Request) are listed under the Tables module.

Add Fields To Book Table

1. Open ServiceNow.

2. Click on All → search for Tables.

3. Select System Definition → Tables.

4. From the list, open the table `Book(_book)`.

5. Scroll down to the Columns (Dictionary) related list.

6. Click on New to add a new field.

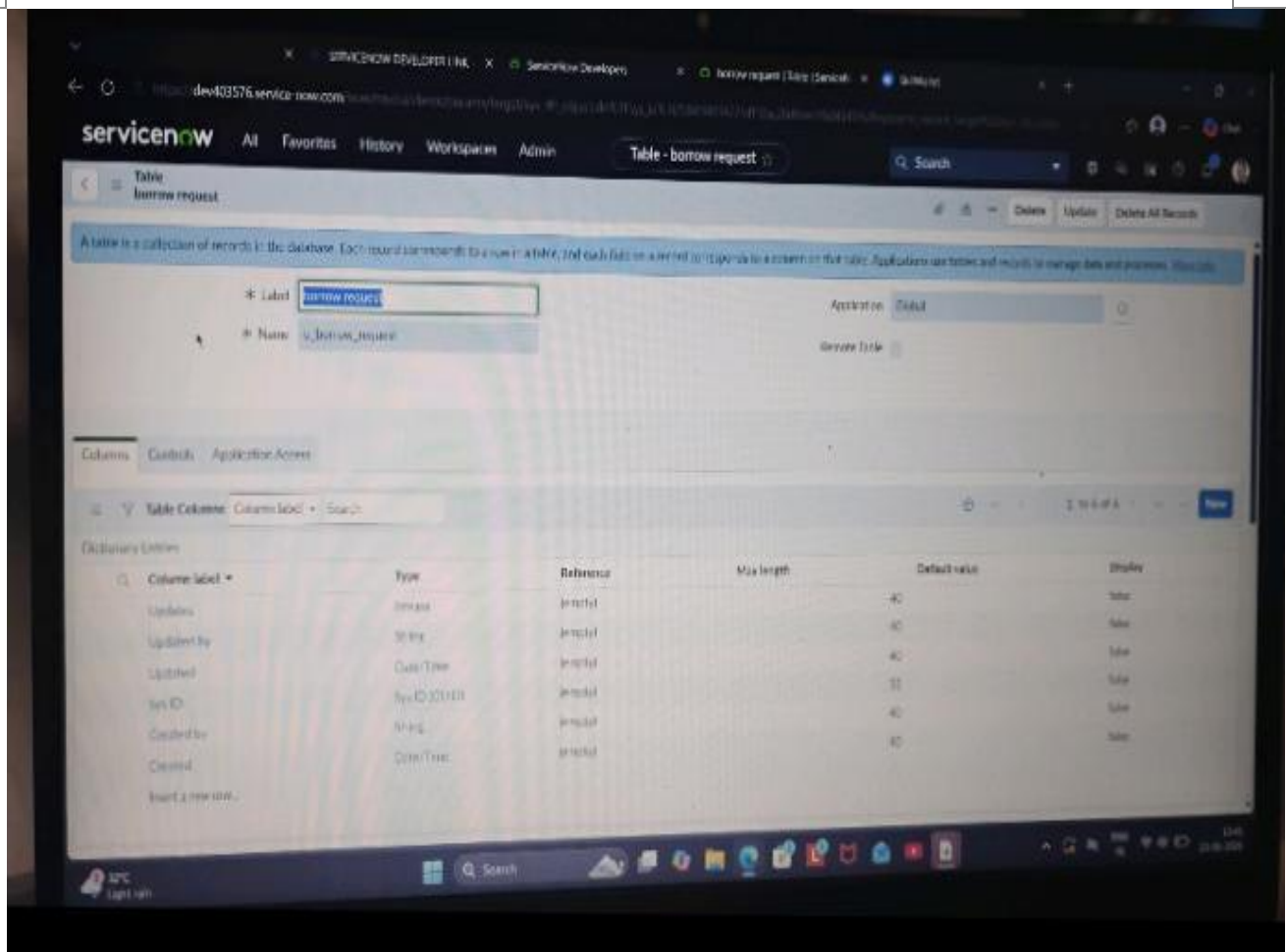
7. Enter Column Label as Title, select *Type= String → click Submit.

8. Again, click on New. Enter Column Label as Author, select *Type= String → click Submit.

9. Again, click on New. Enter Column Label as ISBN, select *Type= String → click Submit.

10. Again, click on New. Enter Column Label as Status, select *Type= Choice → click Submit.

11. Open the Status field you just created.

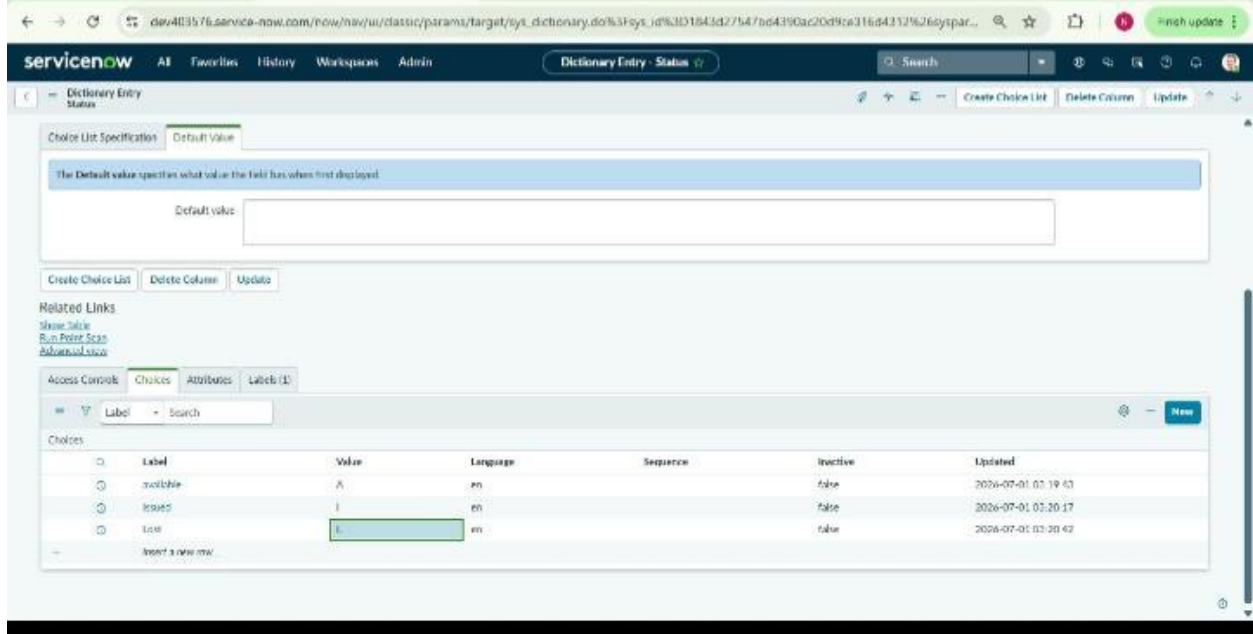


12. In the Choices related list, click New and add values: Available, Issued, Lost.

13. Set the Default Value of Status = Available.

14. Click Update to save.

15. Verify that the Book form now has fields: Title, Author, ISBN, Status.



Add Fieldsto Borrow Request table

1. Open ServiceNow.
2. Click on All → search for Tables.
3. Select System Definition → Tables.
4. From the list, open the table Borrow Request (u_borrow_request).
5. Scroll down to the Columns (Dictionary) related list.
6. Click on New to add a new field.
7. Enter Column Label as Requested By, select Type = Reference, set Reference = User [sys_user] → click Submit.
8. Again, click on New. Enter Column Label as Book, select Type = Reference, set Reference = Book [u_book] → click Submit.
9. Again, click on New. Enter Column Label as Request Date, select Type = Date/Time → click Submit.
10. Again, click on New. Enter Column Label as Status, select Type = Choice → click Submit.
11. Open the Status field you just created.

servicenow All Favorites History Workspaces Admin Table - borrow request15 Search

Table borrow request15 Delete Update Delete All Records

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label borrow request15 Application Global

* Name u_borrow_request15 Remote Table

Columns Controls Application Access

Table Columns Type Search 1 to 10 of 10 New

Column label	Type	Reference	Max length	Default value	Display
status	Choice	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Created	Date/Time	(empty)	40		false
request date	Date/Time	(empty)	40		false
Updates	Integer	(empty)	40		false
book	Reference	User(sys_user)	32		false
requested by	Reference	User(sys_user)	32		false
Created by	String	(empty)	40		false

12. In the Choices related list, click New and add values: Requested, Approved, Rejected, Returned.

13. Set the Default Value of Status = Requested.

14. Click Update to save.

15. Verify that the Borrow Request form now has fields: Requested By, Book, Request Date, Status.

servicenow All Favorites History Workspaces Admin Table - borrow request15 Search

Table borrow request15 Delete Update Delete All Records

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label borrow request15 Application Global

* Name u_borrow_request15 Remote Table

Columns Controls Application Access

Table Columns Type Search 1 to 1 of 1 New

Dictionary entries: Column label starts with request date

Column label	Type	Reference	Max length	Default value	Display
request date	Date/Time	(empty)	40		false
book	Reference		40		false
requested by	Reference		40		false

Insert a new row...

Delete Update Delete All Records

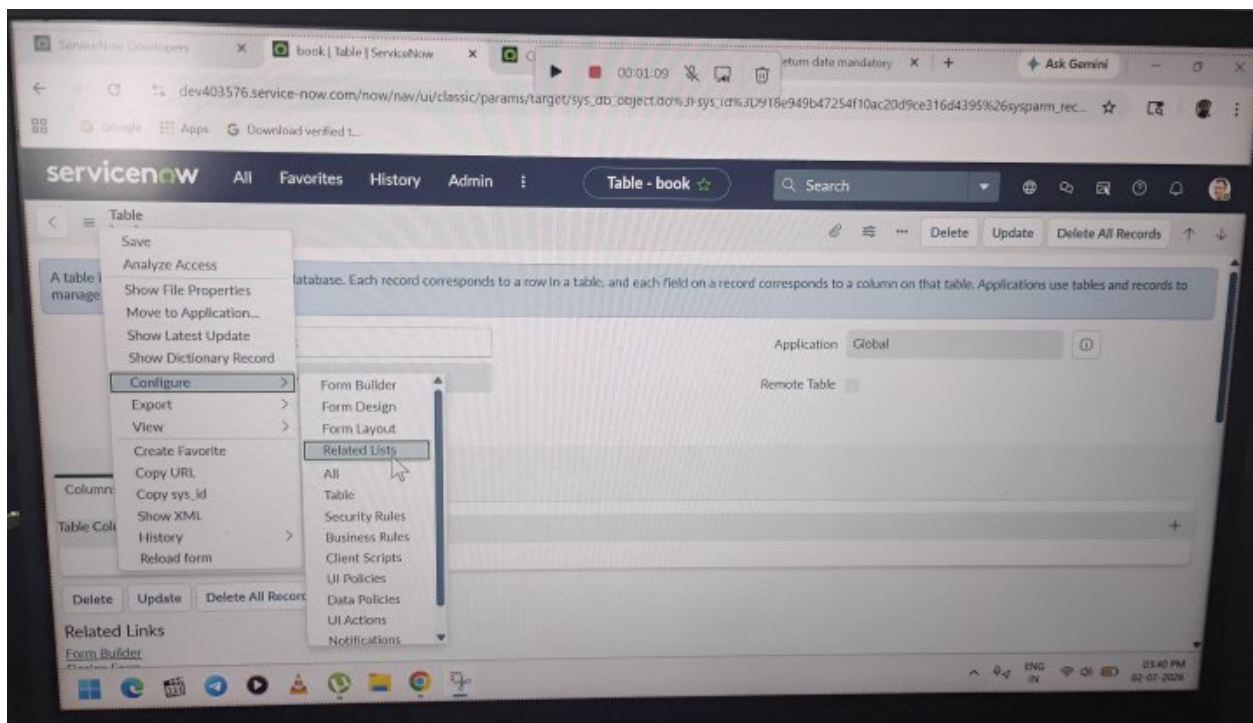
Related Links

[Form Builder](#)

Configure Reference Qualifier (Prevent Requesting Already-Issued Books)

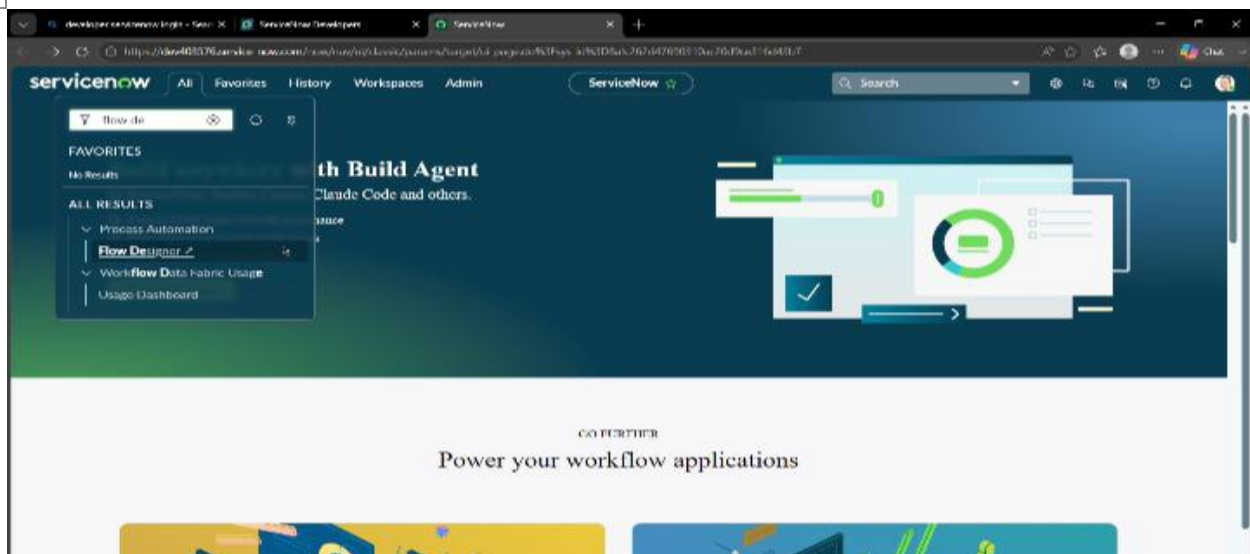
1. Open ServiceNow.

2. Click on All → search for Tables.
3. Select System Definition → Tables.
4. From the list, open the table Borrow Request (u_borrow_request).
5. Scroll down to the Columns (Dictionary) related list.
6. Open the field Book (Reference field to Book [book]).
7. In the field form, scroll to the Advanced View (click the gear if needed).
8. Locate the Reference Qualifier section.
9. Select Simple type.
10. In the Reference Qualifier Condition, set: Status is Available
Click Update to save the field.

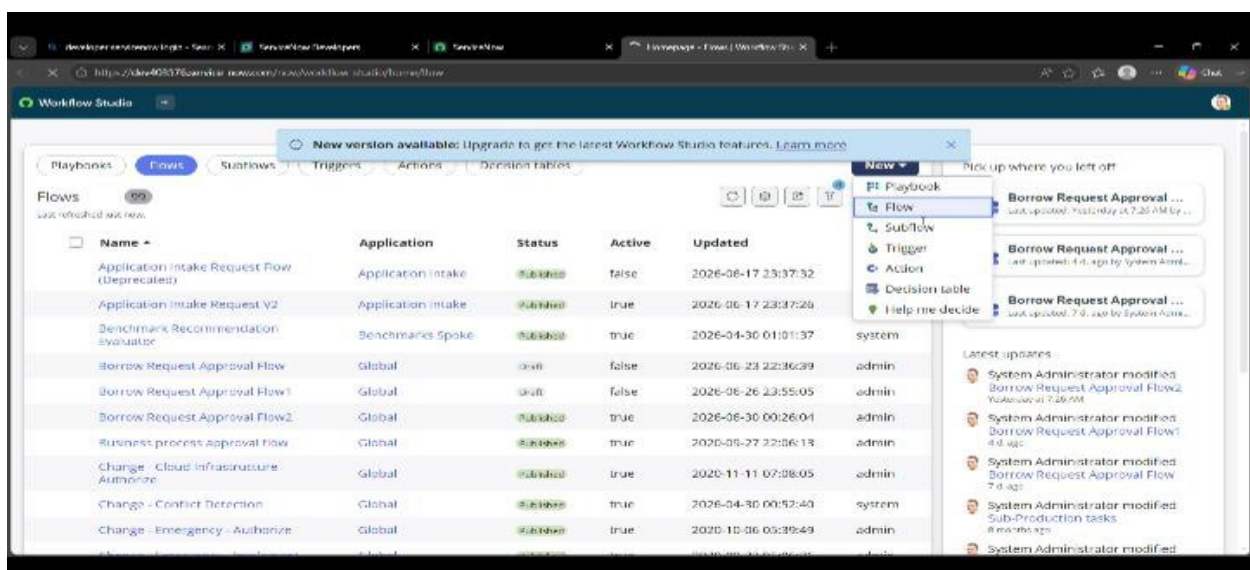


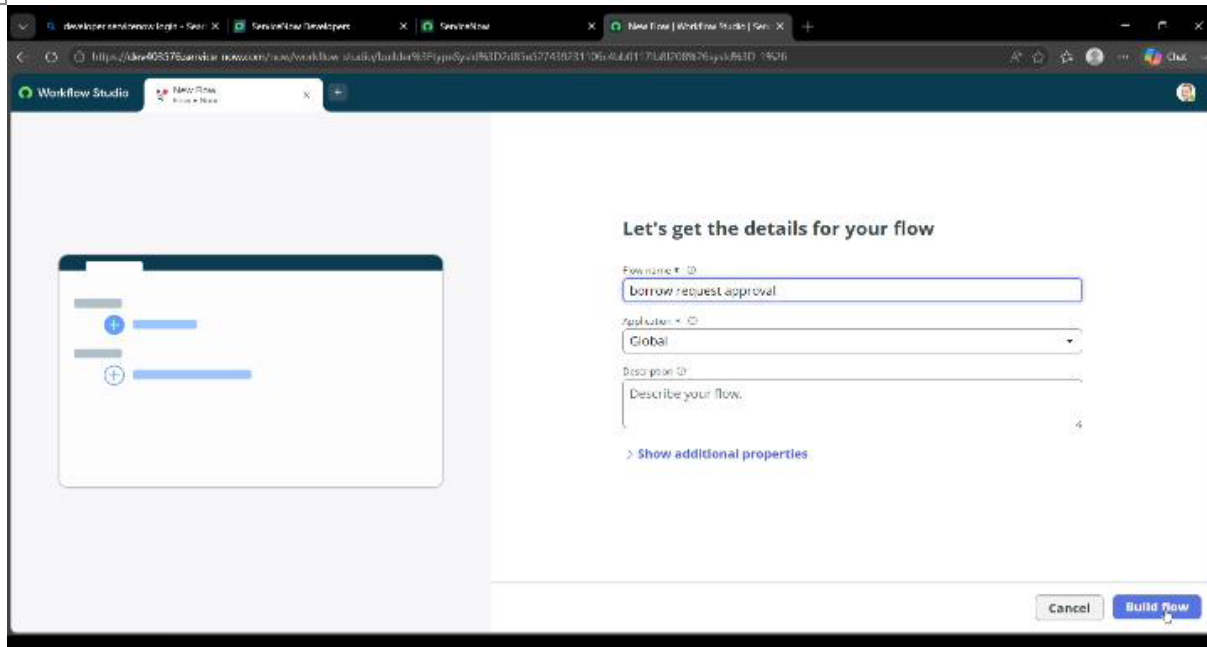
Create Flow in Flow Designer

1. Open ServiceNow.
2. Click on All → search for Flow Designer.
3. Select Flow Designer and click New → Flow.



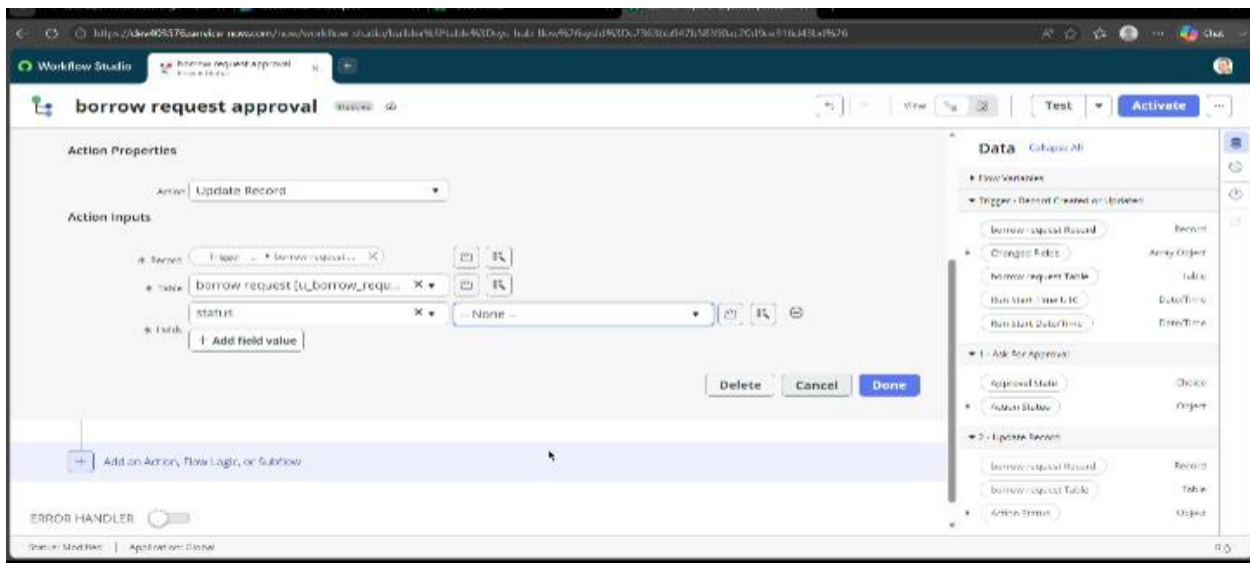
4. Enter Name as Borrow Request Approval Flow → click Submit.





2.Add Trigger (When to Run the Flow)

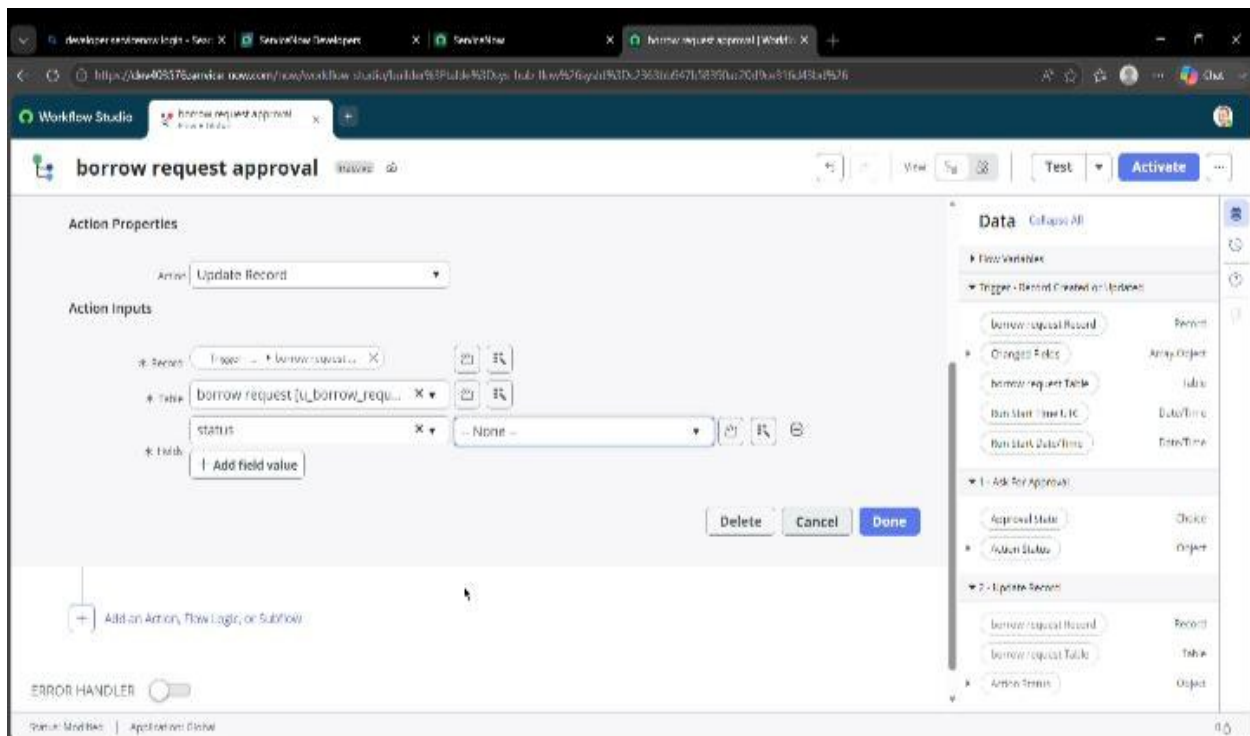
- 1.In the Flow Designer canvas, click + Add Trigger.
- 2.Select Created or Updated.
- 3.Choose Table= Borrow Request [u_borrow_request].
- 4.Add condition: Status= Requested
- 5.Click Done.



3.Add Action – Approval

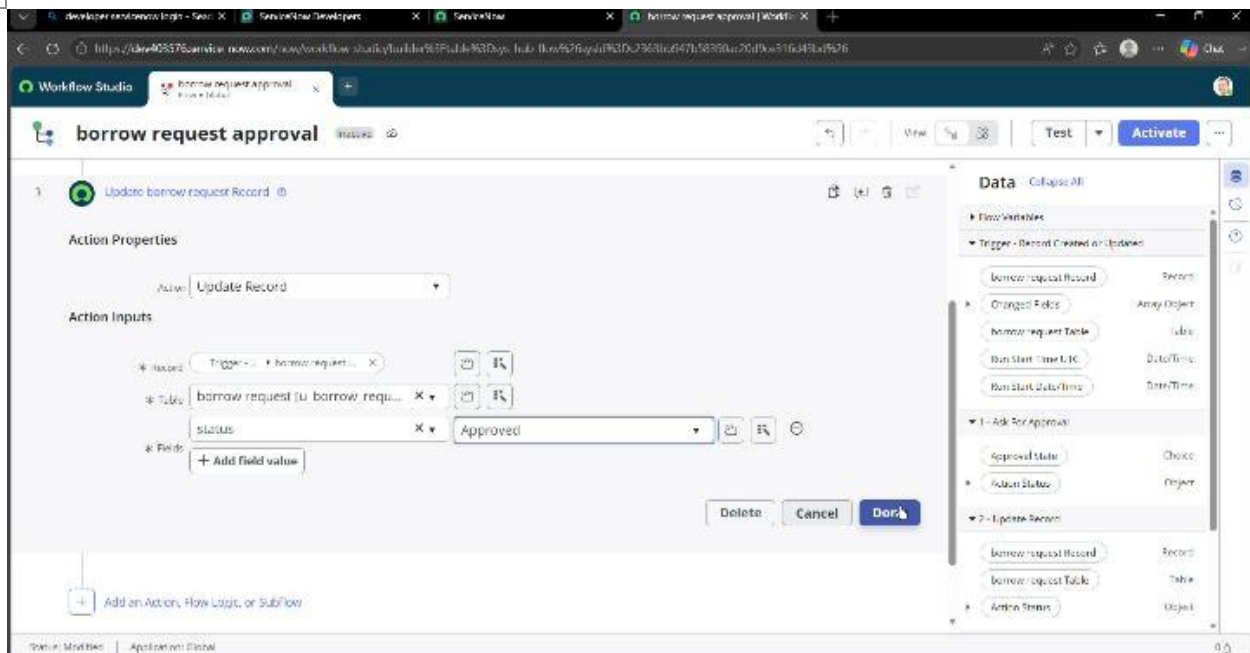
- 1.Click on + Add Action.

2. Search for Ask for Approval.
3. Set Approvers = Role: librarian.
4. Approval conditions:
5. On Approve → continue flow.
6. On Reject → set Status = Rejected.
7. Click Done.



4. Add Action – Update Record (Issue Book)

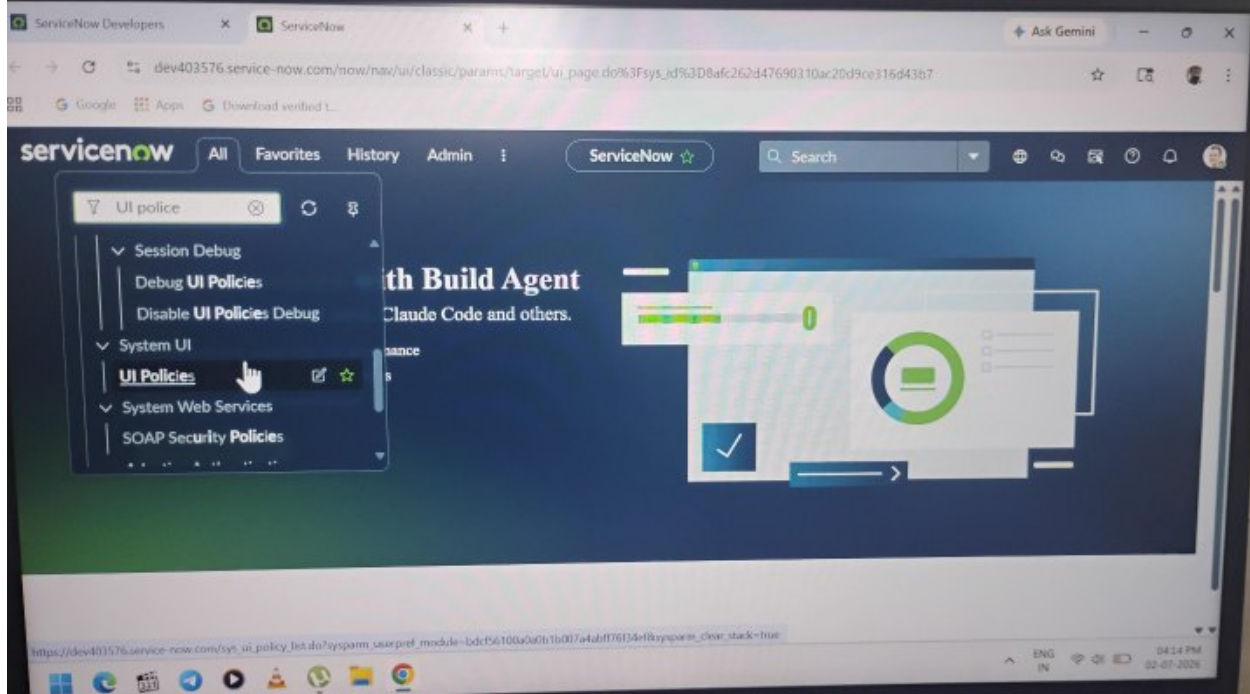
1. Click on + Add Action.
2. Search for an Update Record.
3. Choose Table = Booku[_book].
4. Condition: Sys ID = Current.Book (the one from Borrow Request).
5. Set Status = Issued.
- 6.



Click Done.

5.Add Action – Send Email (Notification)

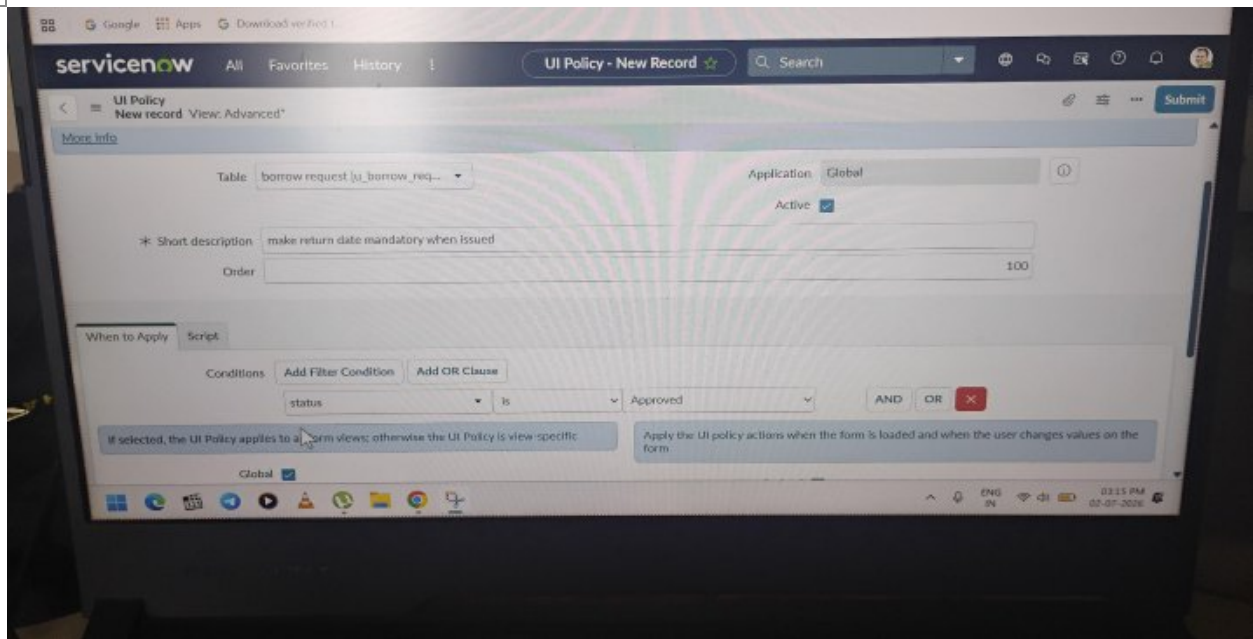
- 1.Click on + Add Action.
- 2.Search for Send Email.
- 3.Set To = Requested By (Borrow Request → Requested By field).
- 4.Subject: Your Borrow Request hasbeen Approved.
- 5.Body: The book [Book.Title] hasbeen issued to you. Please collect it from the library.
- 6.Click Done



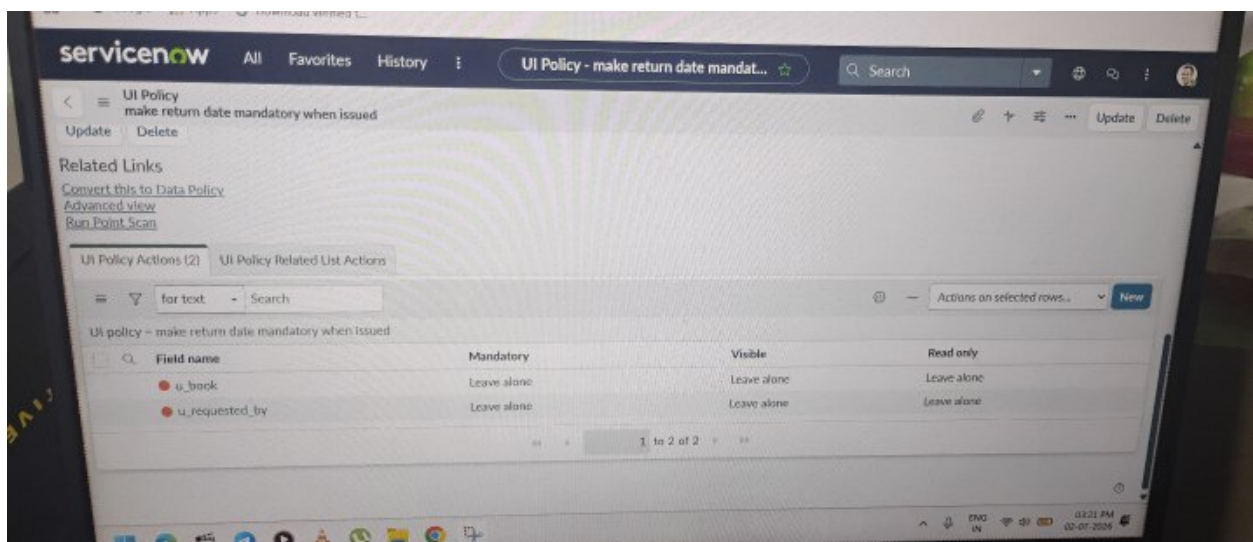
4. Click on New.

5. Enter Table = Borrow Request [u_borrow_request].

6. Enter Short Description = Make Return Date mandatory when Issued.
Submit the record.



7. Reopen the created UI Policy record.
8. In the Conditions section, set: Status = Issued.
9. In the UI Policy Actions related list, click New.
10. Select Field = Return Date.
11. Set Mandatory = True.
12. Set Visible = True.
13. Submit the UI Policy Action.



Access Control for Book Table

1. Open ServiceNow.
2. Click on All → search for Access Control.
3. Select System Security → Access Control (ACL).
4. Click New.
5. Select Table = Booku[book].
6. Choose Operation = read.
7. Set Requires role = student, librarian.

8.

The screenshot shows the 'Access Control - New Record' form in the ServiceNow interface. The form is for creating a new Access Control Record. The 'Type' is set to 'record' and the 'Operation' is set to 'read'. The 'Application' is set to 'Global'. The 'Active' checkbox is checked. The 'Decision Type' is set to 'Allow If'. The 'Protection policy' is set to 'None'. The 'Name' field is empty. The 'Description' field is empty. Below the form, there is a 'Conditions' section with a warning message: 'Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.' The 'Requires role' section is also visible, with a 'Role' field and a 'Insert a new role' button.

Submit.

9. Again, click New.
10. Select Table = Booku[book].
11. Choose Operation = write.
- 12.

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

Type: record

Operation: write

Admin overrides: ☒

Decision Type: Allow if

Protection policy: None

Application: Global

Active: ☒

Advanced: ☐

Name:

Description:

Conditions

Access Control Rules have two decision types, and those types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.

2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

Requires role

Role
Insert a new role...

Set Requires role = librarian.

13. Submit.

14. Again, click New.

15. Select Table = Book [book].

16. Choose Operation = delete.

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

Type: record

Operation: delete

Admin overrides: ☒

Decision Type: Allow if

Protection policy: None

Application: Global

Active: ☒

Advanced: ☐

Name:

Description:

Conditions

Access Control Rules have two decision types, and those types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.

2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

Requires role

Role
Insert a new role...

17. Set Requires role = librarian.

18. Submit.

Access Control for Borrow Request Table

1. Open ServiceNow.

2. Click on All → search for Access Control.
3. Select System Security → Access Control (ACL).
4. Click New.
5. Select Table = Borrow Request [u_borrow_request].
6. Choose Operation = create.
7. Set Requires role = student.
8. Submit.
9. Again, click New.
10. Select Table = Borrow Request [u_borrow_request].
11. Choose Operation = read.
12. Set Requires role = student, librarian.

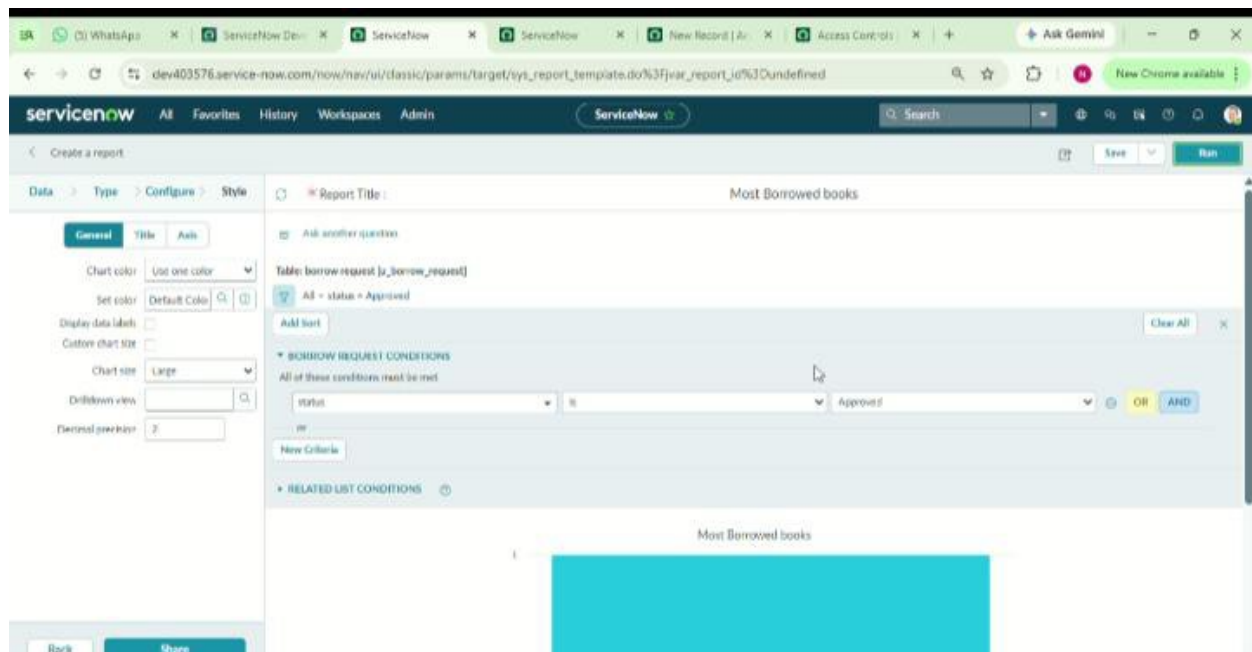
The screenshot shows the ServiceNow 'Access Control - New Record' form. The 'Conditions' section is expanded, showing a list of roles under 'Requires role'. The roles listed are 'student' and 'librarian'. Below this, the 'Security Attribute Condition' section is visible, featuring a search bar and a dropdown menu for selecting a condition. The form is currently in the 'Requires role' step, with a 'Submit' button at the top right.

13. Again, click New.
14. Select Table = Borrow Request [u_borrow_request].
15. Choose Operation = write.
16. Set Requires role = librarian.
17. Submit.
18. Again, click New.
19. Select Table = Borrow Request [u_borrow_request].
20. Choose Operation = delete.
21. Set Requires role = librarian.
22. Submit

Create Report

1. Open ServiceNow.

2. Click on All → search for Reports.
3. Select Reports → View / Run.
4. Click on New.
5. Enter Name = Most Borrowed Books.
6. Select Source Type = Table.
7. Select Table = Borrow Request [u_borrow_request].
8. Choose Type = Bar Chart (or Pie Chart as needed).
9. In the Group by field, select Book.
10. In the Aggregate field, select Count.
11. Add a filter: Status = Approved (to count only issued books).
12. Click Run to preview the report.
13. Adjust chart options (Title, Labels, Legend) as required.
14. Click Save.
15. To publish, click Sharing → Add roles (student, librarian).



Phase Deployment, Documentation & Final Presentation

Troubleshooting

Flow Execution & Logs

Verified end-to-end execution of the Borrow Request Approval Flow using Flow Designer Execution Details. Checked trigger conditions, approvals, and book issuance updates.

Reference Qualifier Issues

Debugged cases where already-issued books were still showing in the Borrow Request form.

Corrected reference qualifier conditions to filter only available books.

UI Policy Behavior

Tested Borrow Request form to ensure Return Date became mandatory when Status = Issued. Verified field visibility and mandatory settings across portal and platform views.

Access Control Validation

Checked that students could only create/read their own Borrow Requests while librarians had full access. Adjusted ACL conditions to restrict unauthorized access.

Report Accuracy

Validated the "Most Borrowed Books" report to ensure only approved/issued requests were counted. Fixed grouping to display correct book titles.

Adherence to Timelines

1. Executed project based on the defined milestones:
2. Creation of Roles
3. Creation of Tables
4. Adding Fields (Book & Borrow Request)
5. Reference Qualifier Implementation
6. Flow Designer Setup (Approval & Issuance)
7. UI Policy Configuration
8. Access Control Setup
9. Report Creation ("Most Borrowed Books")
10. Testing & Validation

Innovation

End-to-End Automation: Automated the book borrowing process by integrating custom tables, Flow Designer, and approvals, eliminating manual tracking of borrow requests.

Optimized Request Handling: Implemented reference qualifiers to ensure only available books can be requested, reducing errors and duplicate borrow entries.

Maintainability & Scalability: Designed the solution using out-of-the-box ServiceNow features like Flow Designer, UI Policies, and ACLs, ensuring easy maintenance and future scalability.

User Experience Enhancements: Applied UI policies to make Return Date mandatory only when a book is issued, keeping forms clean and user-friendly for both students and librarians.

Document Technical Blueprint

Variables

1. Book Title and Author
2. Borrower (auto-populated with current user, editable if borrowing for others)

3. Borrow Date (auto-filled with current date)
 4. Return Date (mandatory only when book is issued)
- Approval Use Cases:
5. Librarian Approval required automatically when a student raises a borrow request.
 6. Auto-approval for requests made by librarian role (for testing or internal usage).
- Fulfillment Task Use Cases:
8. Catalog tasks created for the Librarian group to issue the book.
 9. Automatic update of Borrow Request status to "Issued" once approved.
 10. Automatic update of Borrow Request status to "Returned" when the book is marked as returned.

Document Setup Manual

Provided step-by-step manual for recreating the solution in a new Personal Developer Instance (PDI):

- Role creation (Librarian, Student)
- Table creation (Book, Borrow Request)
- Field configuration for each table
- Reference qualifier setup for available books
- Flow Designer configuration (Approval → Issue → Return)
- UI Policy setup for mandatory Return Date
- Access Control rules for students and librarians
- Report creation ("Most Borrowed Books")
- Test data submission performed to validate borrow/return flow
- Screenshots and order of operations included for clarity

Project Demo Video Planning

Demo includes:

- Portal walkthrough: Home → Library Services → Submit Borrow Request
- Live book borrow request submission by a student
- Dynamic approval flow handled by librarian
- Email notification triggers for request approval and issuance
- Request and book status tracking (Requested → Approved → Issued → Returned)
- Includes narration for each step and role-based perspectives (Student, Librarian)

Visual Demonstration

Captured:

- Form UI behaviour (e.g., Return Date becomes mandatory only when a book is issued)
- Approval records in the Borrow Request timeline (Student request → Librarian approval → Issued)
- Flow logs showing the end-to-end execution path (Request submitted → Approved → Issued → Returned)
- Annotated visuals with labels and highlights to explain each stage clearly

Explanation Clarity

Explanation Clarity

Documentation and presentation follow logical, sequential structure:

What → Why → How → Outcome

Each step includes:

Clear headings for every configuration (Roles, Tables, Flow, UI Policy, etc.)

Labelled screenshots of ServiceNow setup and execution

Explanatory notes for easy replication in a new PDI

Realism & Quality

Workflow mimics real-world library operations, including:

○ Role-based approvals (Student requests → Librarian approves)

○ Data validations (book availability check, mandatory return date)

Solution tested with realistic use cases and sample data (multiple students requesting the same book, overdue returns).

Scalability & Future Plan

Future enhancements considered:

Auto-notifications for overdue books and reminders via email/SMS

Integration with external library systems or digital catalog APIs

Enhanced reporting dashboards (borrow frequency, student activity, overdue analysis)

Expansion to other academic services (lab equipment requests, study room booking, digital resource access)

Project Conclusion

The Library Management Project enhances book borrowing and administration in ServiceNow through structured roles, automated approval flows, and workflow-driven processing. Students are able to request books seamlessly, while librarians manage approvals and issuance with built-in validations. Clear status updates and reporting improve visibility and control, and the automated design ensures efficiency, reliability, and scalability for future academic service extensions.

Functional Requirement	User story	No of Activity	Team members
Requirement Analysis & Planning	As a project team, I want business objectives & functional scope defined, so the project direction is clear.	3	B.Chandrakala
	As a BA, I want stakeholder mapping & execution roadmap set, so planning is complete.	1	B.Deepika
Backend Development & Configurations	As a developer, I want roles created & tables set up, so the backend structure is ready.	2	B.Chandrakala
	As a developer, I want fields added to Book & Borrow Request tables with a reference qualifier configured, so already-issued books can't be re-requested.	4	Neeraja Bhyri
Development	As a developer, I want a flow created in Flow Designer, so approval processing is automated.	1	Neeraja Bhyri
	As a developer, I want a UI policy configured, so form	1	B.Renuka

	behavior adapts based on data conditions.		
Data Migration, Testing & Security	As an admin, I want access control set for the Book & Borrow Request tables, so data is secured properly.	3	B.Renuka
Deployment, Documentation & Final Presentation	As an admin, I want access control configured for the Book table & Borrow Request table, so only authorized users can view/modify records.	1	B.Renuka
	As an analyst, I want a report created for tracking borrow/return activity, so usage can be monitored & analyzed.	5	Biddika Neeraja
Project Conclusion	As a team, I want to summarize outcomes & learnings, so the project closes properly.	1	B.Deepika

Advantages and disadvantages

Advantages:

- Unified platform – requests handled through the same portal as IT, HR, and facilities requests
- Strong automation using Flow Designer for approvals, routing, and escalation
- Built-in SLAs ensure timely processing of requests

- Self-service portal with real-time status tracking and notifications
- Reduces manual email/phone communication with library staff
- Powerful reporting and analytics for request volumes, popular items, and turnaround times

Disadvantages

- Not purpose-built for libraries – lacks ISBN/barcode cataloging, circulation, and fines management features
- MARC record support is missing and must be custom-built
- Licensing costs are high compared to dedicated systems like Koha or Evergreen
- Requires ServiceNow development skills (Flow Designer, Glide scripting)
- Customizations may break during platform upgrades, increasing maintenance effort

Project conclusion

Conclusion

The Library Management Project successfully streamlined the process of borrowing and managing books within ServiceNow by integrating role-based access, automated approvals, and dynamic workflows. Students can easily submit borrow requests, while librarians efficiently review, approve, and issue books with proper validations. The solution ensures transparency through status tracking, notifications, and reporting features. By combining automation with usability, the project delivers a scalable foundation that can be extended to broader academic services in the future.

Assign Stories

Filter stories

Unassigned Stories: 0

Members without Stories: 0

Business Objectives Epic: Requirement Analysis & Planning 20m Easy INPROGRESS	Assign to: Bobbie Chandra
Functional Scope Epic: Requirement Analysis & Planning 20m Easy INPROGRESS	Assign to: Bobbie Chandra
Stakeholder Mapping Epic: Requirement Analysis & Planning 20m Easy INPROGRESS	Assign to: Bobbie Chandra
Execution Roadmap Epic: Requirement Analysis & Planning 9h-0m Easy INPROGRESS	Assign to: Demetri Deshpande
Creation of roles Epic: Backend Development & Configurations 10m Easy INPROGRESS	Assign to: Bobbie Chandra
Create Tables Epic: Backend Development & Configurations 10m Easy INPROGRESS	Assign to: Bobbie Chandra
Create Tables Epic: Backend Development & Configurations 10m Easy INPROGRESS	Assign to: Neeraja Shyl
Add Fields to Book table Epic: Backend Development & Configurations 20m Easy INPROGRESS	Assign to: Neeraja Shyl
Add Fields to Borrow Request table Epic: Backend Development & Configurations 20m Easy INPROGRESS	Assign to: Neeraja Shyl
Configure Reference Qualifier (Prevent Requesting Already-Issued Books) Epic: Backend Development & Configurations 20m Easy INPROGRESS	Assign to: Neeraja Shyl
Create Flow in Flow Designer Epic: Development 20m Easy INPROGRESS	Assign to: Neeraja Shyl
Create UI Policy Epic: Development 20m Easy INPROGRESS	Assign to: Bojjanamuruga Renu
Access Control for Book Table Epic: Data Migration,Testing & Security 20m Easy INPROGRESS	Assign to: Bojjanamuruga Renu
Access Control for Borrow Request Table Epic: Data Migration,Testing & Security 20m Easy INPROGRESS	Assign to: Bojjanamuruga Renu

Configure Reference Qualifier (Prevent Requesting Already-Issued Books)

Epic Backend Development & Configurations

20m

Easy

INFO@EBC30

Assign to:

 Nwanaja Shyl -

Create Flow in Flow Designer

Epic Development

20m

Easy

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Assign to:

 Nwanaja Shyl -

Create UI Policy

Epic Development

20m

Easy

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Assign to:

 Boyinawemuka Iben -

Access Control for Book Table

Epic Data Migration, Testing & Security

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Access Control for Borrow Request Table

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Create Report

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Troubleshooting

Epic Deployment, Documentation & Final Presentation

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Adherence to Timelines

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Innovation

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Document Technical Blueprint

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Document Setup Manual

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Project Demo Video Planning

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Visual Demonstration

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Explanation Clarity

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Conclusion

Epic Project Conclusion

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Assign to:

 Berna Deepika -